DSA06 – DATA HANDLING AND VISUALIZATION

TABLEAU INTRODUCTION ACTIVITY (25 MARKS)

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Submission detail	Value
Dataset	D_13.csv.xlsx (uploaded data file; 48 records)
Tool	Tableau Public (with R/ggplot2 code for verification and comparison)
Period	January–December 2025
Regions	Andhra Pradesh, Gujarat, Karnataka, West Bengal
Fields	Month, Region, Sales, Profit, Customers, Discount_pct

Purpose: To connect and inspect the supplied dataset, create regional and time-series visualizations, investigate discount–profit relationships, build a map and dashboard, and compare Tableau with R.

Dataset overview and preparation

Field	Data type	Role in Tableau	Description
Month	Date	Dimension (continuous Month for time series)	Month-start date in 2025
Region	String	Dimension	Indian state/region
Sales	Number (decimal)	Measure	Sales in Rs lakh
Profit	Number (decimal)	Measure	Profit in Rs lakh; negative values are possible
Customers	Whole number	Measure	Customer count
Discount_pct	Number (decimal)	Measure	Discount percentage

There are 48 rows: 12 months × 4 regions. In Tableau, Month and Region are dimensions; Sales, Profit, Customers and Discount_pct are measures by default. For the scatterplot, use Analysis → uncheck Aggregate Measures so each row becomes an individual mark.

T1. Connect and check [3 marks]
Procedure in Tableau Public:

1. Open Tableau Public and select Connect → To a File → Text file. Choose the supplied D_13 file. If it is provided as an Excel wrapper (.xlsx), choose Microsoft Excel instead; if the instructor expects CSV, save/export the sheet as D_13.csv first.

2. On the Data Source page, verify that Month is Date and the other fields have sensible types. If Month is not Date, click its data-type icon and select Date.

3. Open a worksheet. Drag Sales and Profit to Text, or create a text table with SUM(Sales) and SUM(Profit). Keep the default SUM aggregation.

4. Check the data grid and confirm 48 records and no missing rows.

T1 result	**Answer from supplied file**
Number of rows	48
Dimensions	Month, Region
Measures	Sales, Profit, Customers, Discount_pct
Total Sales	Rs 4539.1 lakh
Total Profit	Rs 430.3 lakh

Interpretation: Across all four regions and all twelve months, total sales were Rs 4539.1 lakh and total profit was Rs 430.3 lakh.

T2. Bar chart: Sales by Region [4 marks]
Tableau build steps:

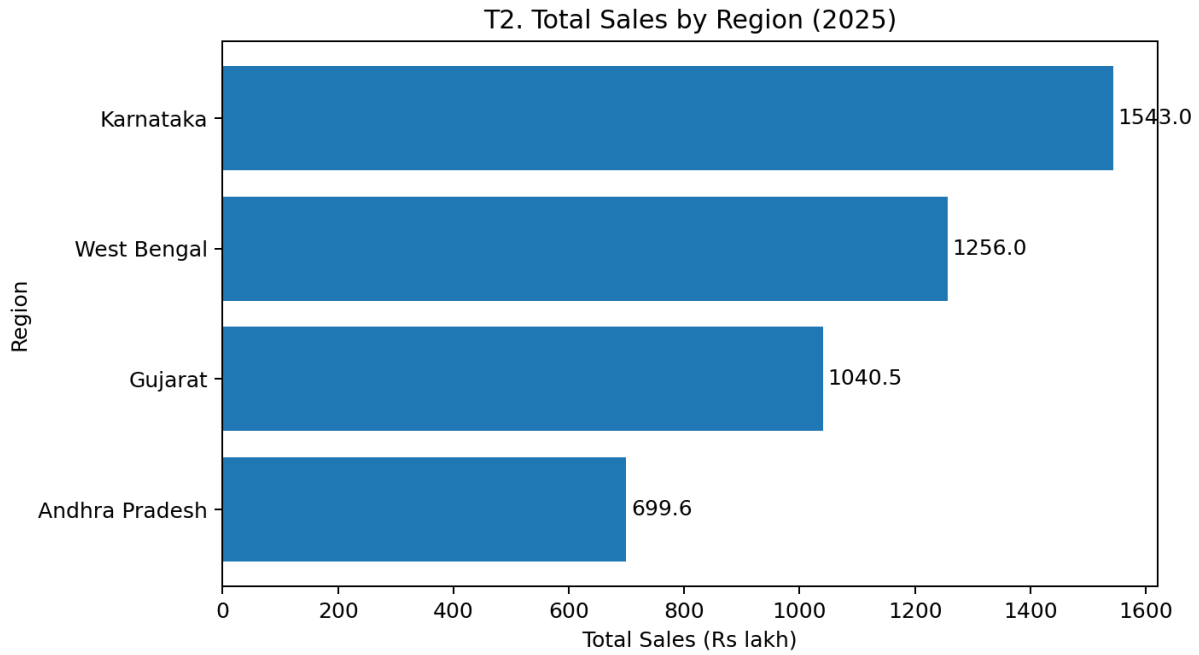
1. Create a new worksheet named “Sales by Region”.

2. Drag Region to Rows and Sales to Columns. Ensure the pill is SUM(Sales).

3. Choose Bar in Show Me (or use the bar mark type). Sort descending by SUM(Sales).

4. Drag SUM(Sales) to Label on the Marks card and enable Show mark labels.

5. Set the title to “Total Sales by Region — 2025”; label the horizontal axis “Sales (Rs lakh)” and the vertical axis “Region”.



Sales rank	Region	Total Sales (Rs lakh)	Share of total sales
1	Karnataka	1543.0	33.99%
2	West Bengal	1256.0	27.67%
3	Gujarat	1040.5	22.92%
4	Andhra Pradesh	699.6	15.41%

Top region: Karnataka, with Rs 1543.0 lakh sales. Its share of total sales is 33.99% = (1543.0 ÷ 4539.1) × 100.

Lowest region: Andhra Pradesh, with Rs 699.6 lakh sales.

T3. Line chart, moving average and trend [5 marks]

Tableau build steps:

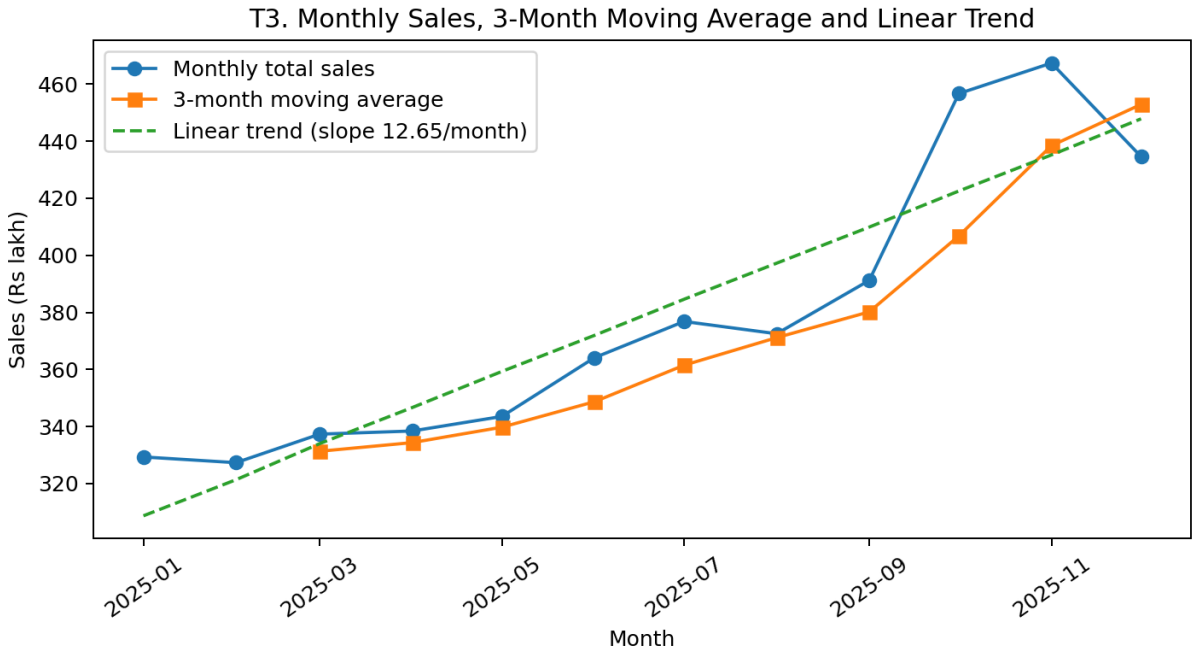
1. Create a worksheet named “Monthly Sales Trend”. Drag Month to Columns and select the continuous Month date level (green pill). Drag Sales to Rows; leave it as SUM(Sales).

2. Choose Line in the Marks card. The view should show 12 monthly totals, each combining the four regions.

3. Add a linear trend: open Analytics and drag Trend Line → Linear onto the view. Alternatively use Analysis → Trend Lines → Show Trend Lines.

4. Add a 3-month moving average: duplicate SUM(Sales) on Rows (Ctrl-drag or duplicate the pill), then right-click the second SUM(Sales) pill → Quick Table Calculation → Moving Average. Edit the table calculation and set Previous values = 2, Next values = 0, Compute Using = Month. This gives a trailing 3-month average including the current month.

5. If the moving average overlays the raw line, use a dual axis only if desired and clearly label both lines; otherwise put the two measures on separate panes. Format the dates as month labels.



Required result	Answer
Peak month	November 2025 — Rs 467.4 lakh
3-month moving average at June	Rs 348.67 lakh = (April + May + June sales) ÷ 3 = (338.4 + 343.5 + 364.1) ÷ 3
Linear trend slope	Approximately +Rs 12.65 lakh per month
Trend interpretation	The fitted linear trend rises by about Rs 12.65 lakh per month on average across the year.

For the trend model, right-click the trend line and choose Describe Trend Model. With Month represented as the sequential monthly period (1–12), the fitted line is

approximately Sales = 296.02 + 12.65 × MonthIndex. The slope is the coefficient of MonthIndex. Tableau's exact display can vary with date treatment; report the slope shown by your own Describe Trend Model dialog if it differs.

Month	Total Sales (Rs lakh)	Trailing 3-month MA (Rs lakh)
Jan 2025	329.3	—
Feb 2025	327.3	—
Mar 2025	337.3	331.30
Apr 2025	338.4	334.33
May 2025	343.5	339.73
Jun 2025	364.1	348.67
Jul 2025	376.8	361.47
Aug 2025	372.5	371.13
Sep 2025	391.2	380.17
Oct 2025	456.7	406.80
Nov 2025	467.4	438.43
Dec 2025	434.6	452.90

T4. Scatterplot: Discount against Profit [4 marks]
Tableau build steps:

1. Create a worksheet named "Discount vs Profit". Drag Discount_pct to Columns and Profit to Rows.

2. Choose Circle on the Marks card. Drag Region to Color so each state has a different colour.

3. Select Analysis → uncheck Aggregate Measures. Verify that the view contains 48 points, one per original row.

4. Open Analytics and drag Trend Line → Linear into the view. Keep one overall linear trend line. Do not add Region to Detail unless you want to preserve each record separately; with aggregate measures off, each record is already represented.

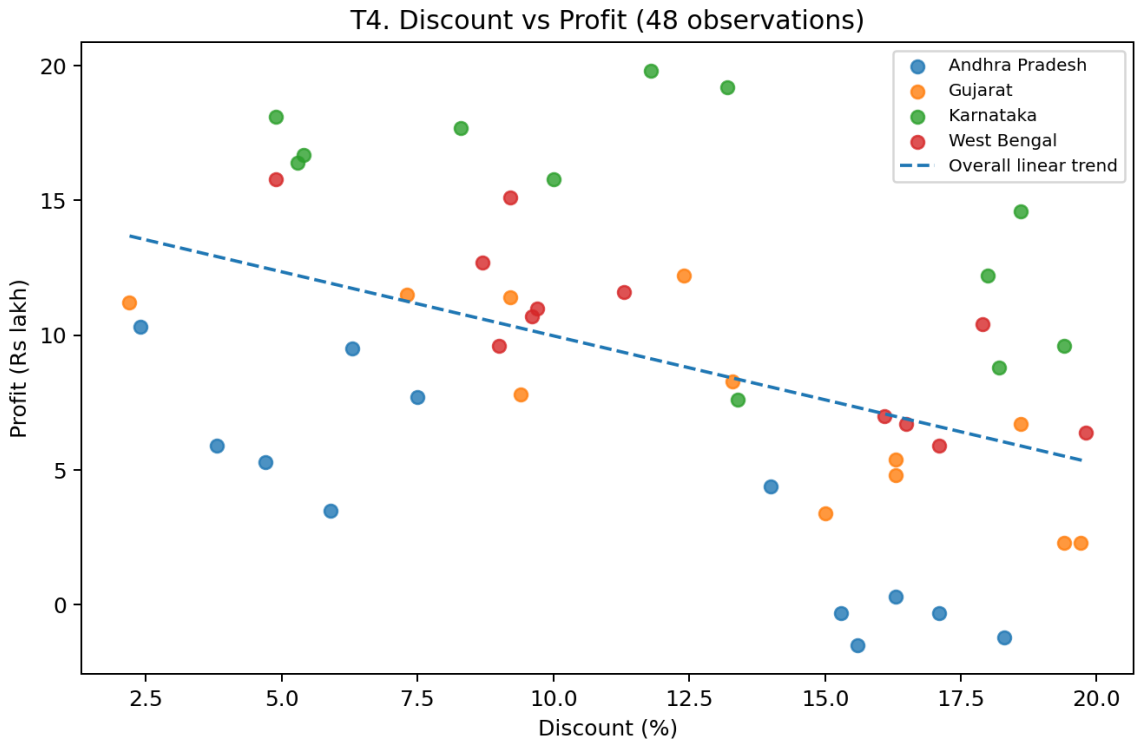
5. Title the chart "Discount (%) vs Profit (Rs lakh)"; label both axes.

```
# Install packages once if required

install.packages(c("readr", "dplyr", "ggplot2", "lubridate", "slider"))
```

```
library(readr)
library(dplyr)
library(ggplot2)
library(lubridate)
library(slider)
# Load dataset
df <- read_csv("D_13.csv", show_col_types = FALSE)
df$Month <- as.Date(df$Month)
# Linear regression: Profit against Discount_pct
fit <- lm(Profit ~ Discount_pct, data = df)
# Regression results, slope and R-squared
summary(fit)
slope <- coef(fit)[["Discount_pct"]]
r <- cor(df$Discount_pct, df$Profit)
cat("Slope:", round(slope, 4), "\n")
cat("Correlation r:", round(r, 4), "\n")
cat("R-squared:", round(summary(fit)$r.squared, 4), "\n")
# Scatterplot with one overall trend line
ggplot(df, aes(x = Discount_pct, y = Profit, colour = Region)) +
  geom_point(size = 3) +
  geom_smooth(
    aes(group = 1),
    method = "lm",
    se = FALSE,
```

```
  colour = "black",
  linetype = "dashed"
) +
labs(
  title = "Discount Percentage vs Profit",
  x = "Discount (%)",
  y = "Profit (Rs lakh)",
  colour = "Region"
) +
theme_minimal()
```



Statistic	Computed result
Slope of Profit on Discount_pct	-0.475 Rs lakh profit per 1 percentage-point increase in discount
Pearson correlation r	-0.455
Relationship	Negative, moderate linear association in this dataset

Interpretation: In these observations, higher discounts tend to be associated with lower profit; the fitted slope is about -0.475 lakh per additional discount percentage point, but this is an association and does not by itself prove that discounting causes lower profit.

Note: The requested r is the signed square root of R-squared. Because the fitted slope is negative, r is negative; the value above is the Pearson correlation for the same 48 paired observations.

T5. Filled map: Profit by Region [3 marks]
Tableau build steps:

1. Ensure Region contains state names and assign Geographic Role → State/Province if Tableau does not recognise them automatically. Because these are Indian states, set the country/region context to India where necessary (Map → Edit Locations).

2. Double-click Region to create a map. If Tableau resolves the states correctly, choose Filled Map in Show Me.

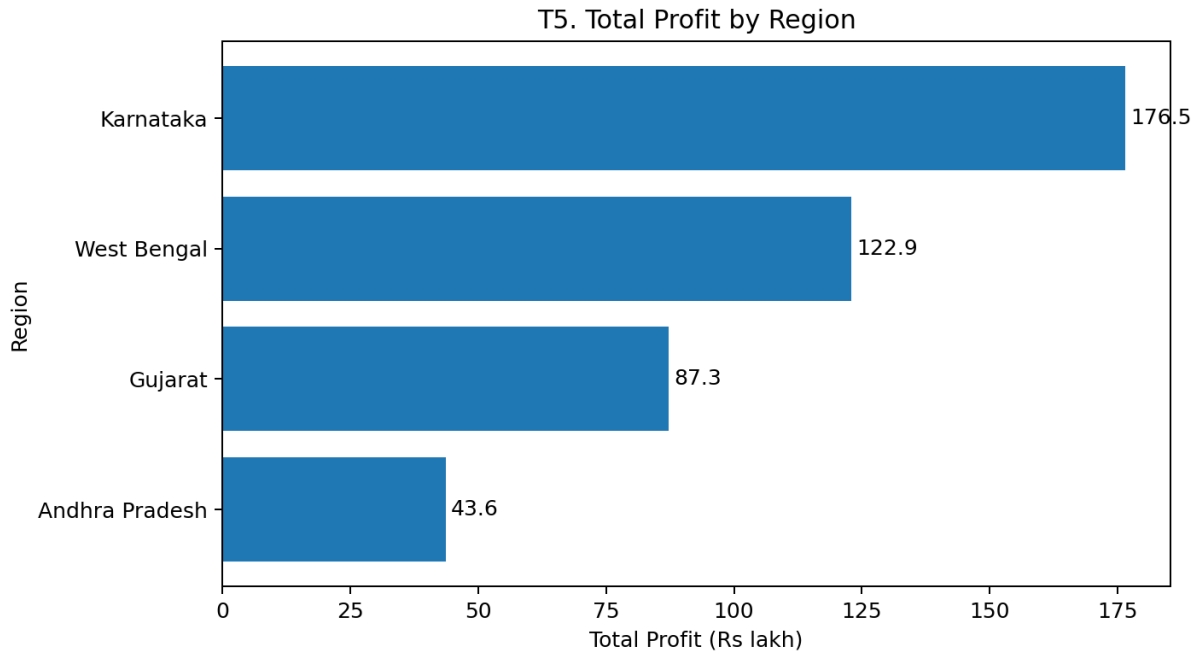
3. Drag SUM(Profit) to Color and optionally SUM(Profit) to Label. Use a diverging palette if negative profit values exist at the row level, but label the view clearly as total regional profit.

4. Title the map “Total Profit by Region — 2025”. If Tableau Public cannot geocode a state, use Map → Edit Locations and set Country/Region to India; do not substitute a different geography.

```
regional_profit <- df %>%
  group_by(Region) %>%
  summarise(
    Total_Sales = sum(Sales, na.rm = TRUE),
    Total_Profit = sum(Profit, na.rm = TRUE),
    Profit_Margin_pct = Total_Profit / Total_Sales * 100,
```

```
    .groups = "drop"
  ) %>%
  arrange(Total_Profit)


print(regional_profit)
cat("Lowest profit region:",
    regional_profit$Region[1], "\n")
cat("Profit margin (%):",
    round(regional_profit$Profit_Margin_pct[1], 2), "\n")
```



Region	Total Profit (Rs lakh)	Total Sales (Rs lakh)	Profit margin
Karnataka	176.5	1543.0	11.44%
West Bengal	122.9	1256.0	9.79%
Gujarat	87.3	1040.5	8.39%
Andhra Pradesh	43.6	699.6	6.23%

Lowest total-profit region: Andhra Pradesh (Rs 43.6 lakh). Profit margin = Profit ÷ Sales × 100 = 43.6 ÷ 699.6 × 100 = 6.23%.

T6. Dashboard with a filter [4 marks]
Dashboard construction:

1. Create a new Dashboard and choose a size such as Automatic or Desktop Browser.

2. Drag the “Sales by Region” bar-chart sheet, “Monthly Sales Trend” line-chart sheet and “Profit by Region” map sheet onto the dashboard. Arrange them so titles and axes remain readable.

3. Click the bar chart and choose Use as Filter (funnel icon), or from the sheet’s drop-down menu choose Use as Filter.

4. Click the SECOND-highest sales bar, West Bengal. The selected dashboard views should filter to West Bengal. Verify the regional totals in a KPI/text table or by hovering over the mark.

5. Publish using Server → Tableau Public → Save to Tableau Public. Sign in, give the workbook a descriptive name, wait for the browser page to open, and copy the public workbook URL into the submission. Do not paste a guessed or unpublished link.

```
regional_summary <- df %>%
  group_by(Region) %>%
  summarise(
    Total_Sales = sum(Sales, na.rm = TRUE),
    Total_Profit = sum(Profit, na.rm = TRUE),
    Profit_Margin_pct = Total_Profit / Total_Sales * 100,
    .groups = "drop"
  ) %>%
  arrange(desc(Total_Sales))
# Display the second-highest sales region
second_highest <- regional_summary %>% slice(2)
```

print(second_highest)

Second-highest sales region	Result
Region	West Bengal
Total Sales	Rs 1256.0 lakh
Total Profit	Rs 122.9 lakh
Profit margin	9.79% = (122.9 ÷ 1256.0) × 100
Published Tableau Public link	Paste your actual published workbook URL here: _____

Expected result after selecting West Bengal: Sales Rs 1256.0 lakh; Profit Rs 122.9 lakh; profit margin 9.79%.

Publishing reminder: the Word document cannot publish a workbook on your account. Complete the dashboard and publish it from Tableau Public, then paste the real URL above.

T7. Compare with R (ggplot2) [2 marks]

```
monthly_sales <- df %>%
 group_by(Month) %>%
 summarise(
  Sales = sum(Sales, na.rm = TRUE),
  .groups = "drop"
 ) %>%
 arrange(Month) %>%
 mutate(
  MonthIndex = row_number(),
  Moving_Average_3M = slide_dbl(
   Sales,
   mean,
   .before = 2,
   .complete = TRUE
```

```
    )
  )
# Calculate linear trend model
trend_model <- lm(Sales ~ MonthIndex, data = monthly_sales)
print(monthly_sales)
print(summary(trend_model))
cat(
  "Trend slope per month:",
  round(coef(trend_model)[["MonthIndex"]], 4),
  "Rs lakh\n"
)
# Plot monthly sales, moving average and linear trend
ggplot(monthly_sales, aes(x = Month, y = Sales)) +
  geom_line() +
  geom_point() +
  geom_line(
    aes(y = Moving_Average_3M),
    linewidth = 1,
    na.rm = TRUE
  ) +
  geom_smooth(
    method = "lm",
    se = FALSE,
    linetype = "dashed"
  ) +
  labs(
    title = "Monthly Sales with Moving Average and Linear Trend",
    x = "Month",
```

```
  y = "Sales (Rs lakh)"
) +
theme_minimal()
```

Difference	Tableau	R with ggplot2
Building the chart	Mostly drag-and-drop using shelves, Marks card and Analytics pane.	Write code that explicitly maps Month to x and Sales to y; geoms and scales are controlled in code.
Trend and moving average	Trend line and moving average can be added through the Analytics pane and Quick Table Calculation UI.	Use functions/packages such as geom_smooth(method = 'lm') and slider/zoo/dplyr to calculate a moving average; the calculation window is explicit in code.

Two clear differences: (1) Tableau uses a visual drag-and-drop interface, whereas R creates the chart through reproducible code. (2) Tableau offers built-in trend lines and quick table calculations, while R requires the analyst to specify the regression and moving-average calculation in code, which is easier to version and rerun exactly.

R code for verification and equivalent charts

Save the dataset as D_13.csv in your R working directory (or change the path below). The following script reproduces the summary values and creates equivalent ggplot2 charts. These R charts are verification aids; the assignment still requires creating the actual Tableau worksheets, dashboard and published link.

```
# DSA06 Tableau Introduction Activity — R verification
# Install once if needed:
# install.packages(c("readr", "dplyr", "ggplot2", "lubridate", "slider"))

library(readr)
library(dplyr)
library(ggplot2)
library(lubridate)
library(slider)

d <- read_csv("D_13.csv", show_col_types = FALSE) |>
  mutate(Month = as.Date(Month))
```

```
# T1: row count, totals and field roles
nrow(d)                        # Expected: 48
sum(d$Sales, na.rm = TRUE)       # Expected: 4539.1
sum(d$Profit, na.rm = TRUE)      # Expected: 430.3
str(d)

# Regional summary for T2, T5 and T6
region <- d |>
  group_by(Region) |>
  summarise(Sales = sum(Sales, na.rm = TRUE),
            Profit = sum(Profit, na.rm = TRUE),
            Customers = sum(Customers, na.rm = TRUE),
            Margin_pct = 100 * Profit / Sales,
            .groups = "drop") |>
  arrange(desc(Sales)) |>
  mutate(Sales_share_pct = 100 * Sales / sum(Sales))
print(region)

# T2: bar chart sorted highest to lowest
ggplot(region, aes(x = reorder(Region, Sales), y = Sales)) +
  geom_col() +
  geom_text(aes(label = sprintf("%.1f", Sales)), hjust = -0.15) +
  coord_flip() +
  labs(title = "Total Sales by Region — 2025",
       x = "Region", y = "Sales (Rs lakh)") +
  expand_limits(y = max(region$Sales) * 1.12) +
  theme_minimal()

# T3: aggregate four regional records to a monthly total
monthly <- d |>
  group_by(Month) |>
  summarise(Sales = sum(Sales, na.rm = TRUE), .groups = "drop") |>
  arrange(Month) |>
  mutate(MA3 = slide_dbl(Sales, mean, .before = 2, .complete = TRUE),
         MonthIndex = row_number())
print(monthly)
lm_month <- lm(Sales ~ MonthIndex, data = monthly)
summary(lm_month)                # Slope is coefficient of MonthIndex

ggplot(monthly, aes(x = Month, y = Sales)) +
  geom_line() + geom_point() +
  geom_line(aes(y = MA3), linewidth = 1, na.rm = TRUE) +
```

```
  geom_smooth(method = "lm", se = FALSE, linetype = "dashed") +
  labs(title = "Monthly Sales, 3-Month Moving Average and Trend",
       x = "Month", y = "Sales (Rs lakh)") +
  theme_minimal()

# T4: one point per row; overall trend line
fit_discount <- lm(Profit ~ Discount_pct, data = d)
summary(fit_discount)            # Coefficient is slope; Multiple R-squared
cor(d$Discount_pct, d$Profit)     # Signed correlation r

ggplot(d, aes(x = Discount_pct, y = Profit, colour = Region)) +
  geom_point() +
  geom_smooth(aes(group = 1), method = "lm", se = FALSE,
              colour = "black", linetype = "dashed") +
  labs(title = "Discount (%) vs Profit (Rs lakh)",
       x = "Discount (%)", y = "Profit (Rs lakh)") +
  theme_minimal()

# T5: profit by region
ggplot(region, aes(x = reorder(Region, Profit), y = Profit)) +
  geom_col() + coord_flip() +
  labs(title = "Total Profit by Region — 2025",
       x = "Region", y = "Profit (Rs lakh)") +
  theme_minimal()

# T6: second-highest sales region
region |> slice(2) |> select(Region, Sales, Profit, Margin_pct)

# T7: The ggplot2 chart is created with code; Tableau's worksheet is created
# using shelves/Marks/Analytics. Both can produce a line + trend + MA.
```

Final answer summary

Task	Key result
T1	48 rows; total Sales Rs 4539.1 lakh; total Profit Rs 430.3 lakh
T2	Top: Karnataka (1543.0 lakh; 33.99% share). Lowest: Andhra Pradesh (699.6 lakh).
T3	Peak: November 2025 (467.4 lakh); June trailing MA = 348.67 lakh; trend slope ≈

	+12.65 lakh/month.
T4	Slope ≈ -0.475; r ≈ -0.455; negative association between discount and profit.
T5	Lowest total profit: Andhra Pradesh (43.6 lakh); margin 6.23%.
T6	Second-highest sales: West Bengal — Sales 1256.0 lakh, Profit 122.9 lakh, margin 9.79%.
T7	Tableau is GUI/drag-and-drop; R uses reproducible code. Trend and moving-average setup differs between UI options and explicit code.

Before submission, confirm the trend slope shown in Tableau's Describe Trend Model, ensure the filled map geocodes Indian states correctly, publish the dashboard to Tableau Public, and replace the blank URL line with the actual public link.